

SIMATS ENGINEERING

ASSESSMENT TOOL II — Visualization Analysis Report

Course: Data Handling and Visualization	Code: DSA0601	CO: CO5
Duration: 180 min	Total Marks: 100	Reg No : 192424226 Name: A Roja

COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods. (BL4)

Activity Tool : Visualization Analysis Report

Weightage: 20%

Code of Conduct:

I A Roja [192424226] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A Roja

Assessment Criteria

Criteria	Visualization Selection and Understanding 25%	Visualization Analysis and Interpretation 25%	Application of Visualization Principles 20%	Organization and Presentation 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Visualization Analysis Report – Assessment Tool**R Programming Practical Questions with Datasets**

Instructions: Answer all five questions. Use R programming to construct the required visualizations, analyze the patterns, compare visualization choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

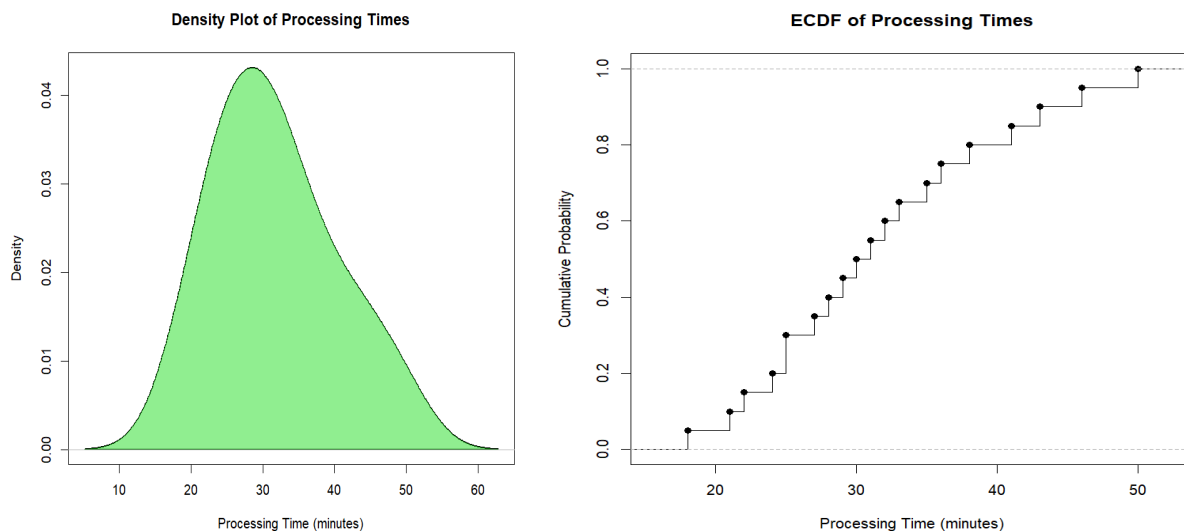
Q1. Comparing Distributions – Histogram, Density and ECDF

A quality-control team records the following processing times (in minutes) for 20 randomly selected tasks. Using R, create a histogram and a density plot for the data. Also construct an empirical cumulative distribution function (ECDF). Compare the three visualizations and explain which aspects of the distribution—center, spread, skewness, and cumulative behavior—are communicated most clearly by each.

Data: Time = 18, 21, 22, 24, 25, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 38, 41, 43, 46, 50 *Expected R tasks: histogram, density estimation, ECDF, appropriate labels/scales, comparison and interpretation.*

Comparing Distributions – Histogram, Density and ECDF**R Code**

```
Time <- c(18, 21, 22, 24, 25, 25, 27, 28, 29, 30,
          31, 32, 33, 35, 36, 38, 41, 43, 46, 50)
hist(Time,
      breaks = 6,
      col = "lightblue",
      border = "black",
      main = "Histogram of Processing Times",
      xlab = "Processing Time (minutes)",
      ylab = "Frequency")
density_time <- density(Time)
plot(density_time,
     main = "Density Plot of Processing Times",
     xlab = "Processing Time (minutes)",
     ylab = "Density",
     lwd = 2)
polygon(density_time,
       col = "lightgreen",
       border = "darkgreen")
plot(ecdf(Time),
     main = "ECDF of Processing Times",
     xlab = "Processing Time (minutes)",
     ylab = "Cumulative Probability",
     verticals = TRUE,
     do.points = TRUE)
mean(Time)
median(Time)
sd(Time)
min(Time)
max(Time)
```



Interpretation

- **Histogram:** Shows the frequency of processing times. Most tasks take around **25–38 minutes** and it clearly shows the center and spread.
- **Density Plot:** Shows the smooth shape of the distribution and helps identify the **slight right skewness**.
- **ECDF:** Shows the cumulative proportion of tasks completed at or below each processing time.
- The processing times range from 18 to 50 minutes, showing a moderate spread.
- The mean is about 31.2 minutes and the median is 30.5 minutes, indicating that the center is around 30–31 minutes.

Conclusion: The histogram, density plot, and ECDF together provide a clear understanding of the processing-time distribution.

Q2. Q-Q Plot and Normality Analysis

A researcher measures the following response times (in seconds). Use R to create a normal Q-Q plot and a histogram/density visualization. Assess whether the observations are reasonably consistent with a normal distribution. Identify any departures from the reference line and explain what they indicate about the distribution.

Data: Response = 42, 45, 47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 63, 66, 70, 76, 85

Expected R tasks: qqnorm()/qqline() or ggplot2 Q-Q plot, histogram/density plot, interpretation of deviations and outliers.

Q-Q Plot and Normality Analysis

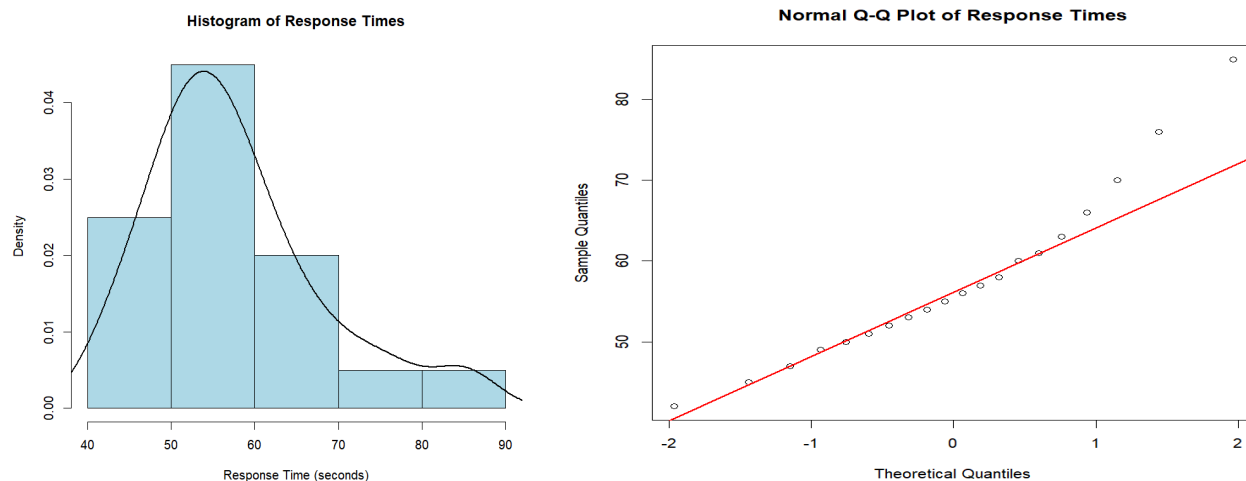
R Code

```
Response <- c(42, 45, 47, 49, 50, 51, 52, 53, 54, 55,
              56, 57, 58, 60, 61, 63, 66, 70, 76, 85)
hist(Response,
      probability = TRUE,
      breaks = 6,
      col = "lightblue",
      border = "black",
      main = "Histogram of Response Times",
      xlab = "Response Time (seconds)",
```

```

    ylab = "Density")
lines(density(Response),
      lwd = 2)
qqnorm(Response,
        main = "Normal Q-Q Plot of Response Times",
        xlab = "Theoretical Quantiles",
        ylab = "Sample Quantiles")
qqline(Response,
        col = "red",
        lwd = 2)
mean(Response)
median(Response)
sd(Response)
min(Response)
max(Response)

```



Interpretation

- The histogram shows that most response times are concentrated around **45–66 seconds**.
- The Q-Q plot points are mostly close to the reference line, indicating that the data is reasonably consistent with a normal distribution.
- The points at the upper end, especially 76 and 85 seconds, move away from the reference line.
- This indicates a slight right-skewness and possible high-value observations.
- Therefore, the data is approximately normal, but it is not perfectly normally distributed.

Conclusion

The Q-Q plot and histogram indicate that the response times are **reasonably close to a normal distribution**, with some deviation in the upper tail due to the higher response times.

Q3. Visualizing Many Distributions at Once

Three groups of students have examination scores recorded below. Create a suitable visualization that compares the distributions of all three groups at once. Use boxplots and/or violin plots, and optionally overlay individual observations. Analyze differences in center, spread, and possible outliers. Explain why the selected visualization is more informative than displaying three separate plots.

Group A = 52,55,57,58,60,61,62,64,65,66,68,70,72,74,78;
 Group B = 45,49,51,53,55,56,58,59,60,62,63,65,67,69,71;
 Group C = 58,60,61,63,64,65,67,68,70,71,72,74,76,80,84

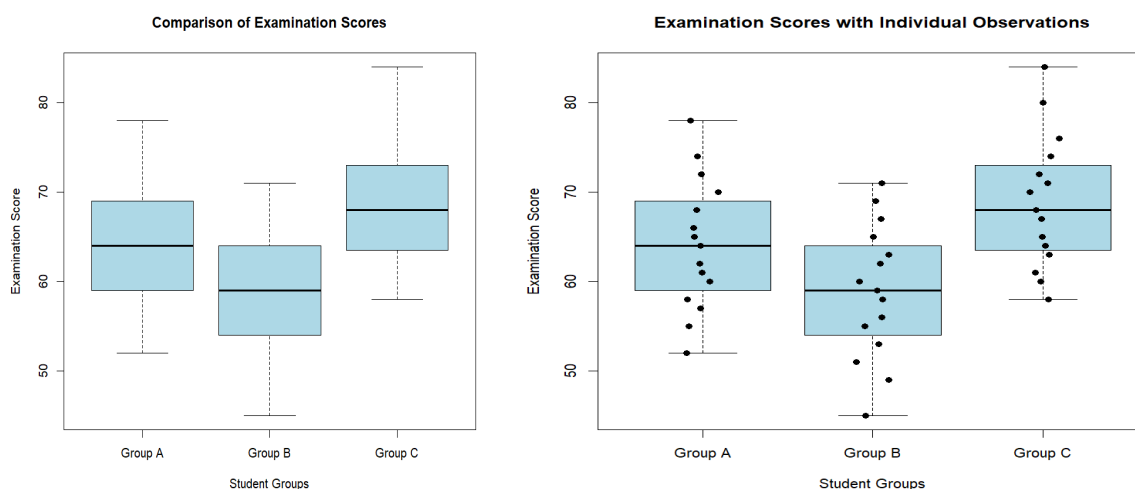
Expected R tasks: data reshaping, grouped boxplot/violin plot, optional jitter, comparison of distributions, interpretation.

Visualizing Many Distributions at Once

R Code

```
Group_A <- c(52,55,57,58,60,61,62,64,65,66,68,70,72,74,78)
Group_B <- c(45,49,51,53,55,56,58,59,60,62,63,65,67,69,71)
Group_C <- c(58,60,61,63,64,65,67,68,70,71,72,74,76,80,84)
Scores <- c(Group_A, Group_B, Group_C)
Group <- factor(c(rep("Group A", 15),
                  rep("Group B", 15),
                  rep("Group C", 15)))

data <- data.frame(Group, Scores)
boxplot(Scores ~ Group,
        data = data,
        col = "lightblue",
        main = "Comparison of Examination Scores",
        xlab = "Student Groups",
        ylab = "Examination Score")
boxplot(Scores ~ Group,
        data = data,
        col = "lightblue",
        main = "Examination Scores with Individual Observations",
        xlab = "Student Groups",
        ylab = "Examination Score")
stripchart(Scores ~ Group,
           data = data,
           method = "jitter",
           vertical = TRUE,
           add = TRUE,
           pch = 19)
aggregate(Scores ~ Group, data = data, mean)
aggregate(Scores ~ Group, data = data, median)
aggregate(Scores ~ Group, data = data, sd)
```



Interpretation

- Group A has a relatively high center, with scores mainly around 60–70.
- Group B has the lowest center and generally lower scores than Groups A and C.

- Group C has the highest center and contains some high scores such as 80 and 84.
- The boxplot allows us to compare the center, spread, and possible outliers of all three groups easily.
- No strong extreme outliers are visible in the groups.
- Showing all three groups in one plot is more informative than using separate plots because the differences between the groups can be compared directly.

Conclusion

The grouped **boxplot with individual observations** is suitable because it clearly compares the **center, spread, and individual score patterns** of all three student groups in a single visualization.

Q4. Relationships and Multivariate Visualization

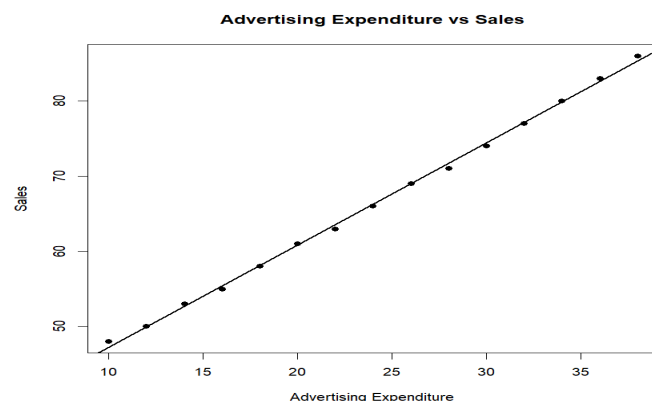
A study records advertising expenditure, website visits, and sales for 15 observations. Create a scatterplot of advertising expenditure versus sales and use an appropriate visual encoding to incorporate website visits. Add a fitted trend line. Analyze the strength and direction of the relationship and explain whether the additional variable improves the interpretation or risks cluttering the visualization.

Advertising = 10,12,14,16,18,20,22,24,26,28,30,32,34,36,38;
 Visits = 120,135,150,165,180,195,210,225,245,260,275,290,305,320,340;
 Sales = 48,50,53,55,58,61,63,66,69,71,74,77,80,83,86

Relationships and Multivariate Visualization

R Code

```
Advertising <- c(10,12,14,16,18,20,22,24,26,28,30,32,34,36,38)
Visits <- c(120,135,150,165,180,195,210,225,245,260,
            275,290,305,320,340)
Sales <- c(48,50,53,55,58,61,63,66,69,71,
           74,77,80,83,86)
plot(Advertising, Sales,
     pch = 19,
     cex = Visits / 150,
     main = "Advertising Expenditure vs Sales",
     xlab = "Advertising Expenditure",
     ylab = "Sales")
model <- lm(Sales ~ Advertising)
abline(model,
       lwd = 2)
cor(Advertising, Sales)
summary(model)
```



Interpretation

- The scatterplot shows a strong positive relationship between advertising expenditure and sales.
- As advertising expenditure increases, sales also increase.
- The points follow an almost straight-line pattern, indicating a strong relationship.
- Website visits are represented using different point sizes. Larger points indicate more website visits.
- The fitted trend line confirms the positive direction of the relationship.
- Adding website visits improves the visualization by providing a third variable, but too many visual encodings could make the graph cluttered and difficult to understand.

Conclusion

The scatterplot with point size for website visits effectively shows the relationship between advertising, website visits, and sales. The data indicates that higher advertising expenditure is associated with higher sales.

Q5. Misleading Visualization and Redesign

A college reports the following pass percentages for seven departments. Create two visualizations: (1) a misleading version using a truncated y-axis and an inappropriate visual emphasis, and (2) a redesigned version using an appropriate baseline, ordered categories, and clear labels. Compare the two plots and explain how the first visualization can exaggerate differences and why the redesigned visualization communicates the data more honestly.

Department = CSE, ECE, EEE, MECH, CIVIL, AIDS, IT;

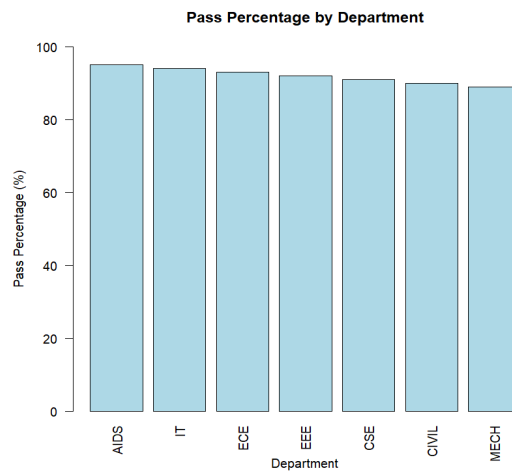
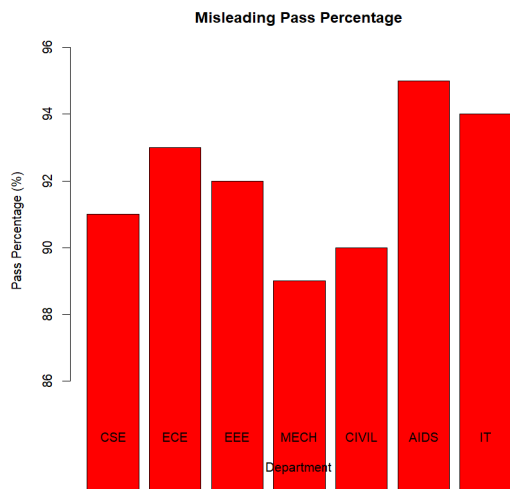
Pass_Percentage = 91,93,92,89,90,95,94

Expected R tasks: bar chart with controlled y-axis, redesigned bar chart, ordering, labels, misleading-axis critique, accessibility and communication analysis.

Misleading Visualization and Redesign

R Code

```
Department <- c("CSE", "ECE", "EEE", "MECH", "CIVIL", "AIDS", "IT")
Pass_Percentage <- c(91, 93, 92, 89, 90, 95, 94)
barplot(Pass_Percentage,
        names.arg = Department,
        ylim = c(85, 96),
        col = "red",
        main = "Misleading Pass Percentage",
        xlab = "Department",
        ylab = "Pass Percentage (%)")
order_data <- order(Pass_Percentage, decreasing = TRUE)
barplot(Pass_Percentage[order_data],
        names.arg = Department[order_data],
        ylim = c(0, 100),
        col = "lightblue",
        main = "Pass Percentage by Department",
        xlab = "Department",
        ylab = "Pass Percentage (%)",
        las = 2)
```



Interpretation

- The misleading chart starts the y-axis at 85% instead of 0%, making small differences between departments appear much larger.
- For example, the difference between 89% and 95% is only 6 percentage points, but the truncated axis can make it look very large.
- The redesigned chart starts the y-axis at 0%, giving a more honest visual comparison.
- The departments are arranged from highest to lowest pass percentage, making comparison easier.
- Clear axis labels and department names improve readability and accessibility.
- The redesigned visualization communicates the actual differences without exaggerating them.

Conclusion

The first chart can mislead viewers by exaggerating small differences through a truncated y-axis. The redesigned chart uses a zero baseline, ordered categories, and clear labels, making the data easier and more honestly understood.